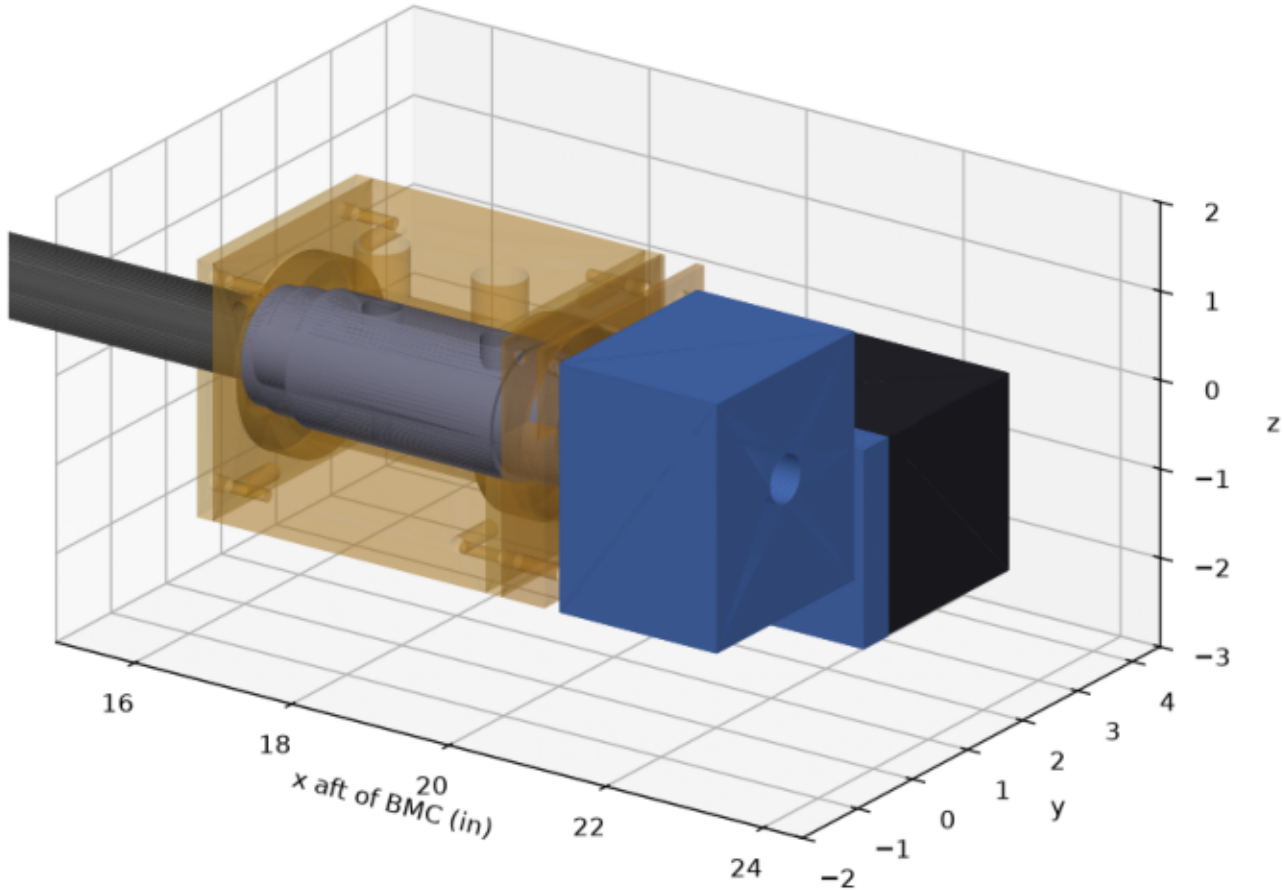


ROLL SPINDLE AT ROLL 0 (CAD RENDER, NTS)

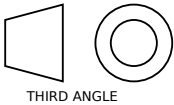
Roll spindle at roll +0 deg: housing/caps/adapter transparent, NMRV030 50:1 + NEMA 23, AS5048A encoder



NOTES:

- 1. PRESS BOTH BEARINGS ONTO THE SPINDLE SEATS; FIT INTO THE HOUSING; CLAMP WITH THE CAPS. SHIM THE REAR CAP FOR .002-.004 AXIAL FLOAT; TIGHTEN THE KM7 LOCKNUT.
- 2. STING FLATS IN THE D-D BORE; 2 BOLTS THROUGH THE TOP WINDOWS AT ROLL 0.
- 3. AS5048A CHIP ON THE SPINDLE AXIS WITHIN 0.25 MM, 1.5 MM AIR GAP.
- 4. BALANCE CABLE EXITS AT X = 20.53: LEAVE A SERVICE LOOP FOR 270 DEG OF TWIST (CALIBRATION MODE).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X ±.03 .XX ±.010
.XXX ±.005 .XXXX ±.0005
ANGLES ±0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 µin Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-201	Spindle, roll	1	4140
2	WT-6C-202	Housing, roll spindle	1	6061-T6
3	WT-6C-203-1	Cap, bearing, front	1	6061-T6
4	WT-6C-203-2	Cap, bearing, rear	1	6061-T6
5	WT-6C-204	Adapter, gearbox	1	6061-T6
6	WT-6C-205	Bracket, roll encoder	1	6061-T6
7	-	Bearing 6907-2RS (35x55x10)	2	bought
8	-	Locknut KM7 + lock washer MB7	1	bought
9	-	Worm gearbox NMRV030 50:1, NEMA 23 input	1	bought
10	-	Stepper motor NEMA 23	1	bought
11	-	Encoder AS5048A board + 6x2.5 diametric magnet	1	bought
12	-	SHCS #8-32 (caps, adapter)	16	bought
13	-	SHCS M6 (gearbox), M3 standoffs (encoder)	1	bought
14	-	Shims .002/.005 (rear cap float)	1	bought

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, ROLL SPINDLE							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-200		A	B	NTS		-	2 OF 35